A Robot Policy Designed as a Weight-Space Artifact Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and exact weight access

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Abstract

Tensor decomposition rewrites a multidimensional table as a small set of low-rank factors. In a neural network, those factors can expose which combinations of inputs and outputs are encoded by the weights. Conventional nonlinear operations make such a table difficult to obtain directly. We introduce χ-VLA, a 20.1M-parameter vision-language-action policy containing only bilinear, linear, and rational learned operations. Its checkpoint therefore specifies an explicit rational tensor program rather than only an opaque executable function. A mechanical audit and local reconstructions verify that claim. We derive an exact weight-based decomposition of each softmax-free attention layer and reconstruct all 12 modules in three checkpoints with worst relative error 2.56×10^{-7}. An $O(d^2)$ reduced Gram avoids materializing the coefficient tensor. All 36 primary block-head tests favor the rank-128 weight-derived subspace over fixed random controls. The full appendix screen finds the effect in $83/96$ pairs and identifies where the signal concentrates across depth. A convolutional instantiation of the same rational model family reaches $93.7\% \pm 0.8\%$ on LIBERO-Object, and its three-checkpoint ensemble reaches 96.2%. We therefore establish a capable model family with exact layer-wise structure derived from trained coefficients. We do not yet provide an exact end-to-end decomposition of the complete policy.

1 Introduction

Neural checkpoints are usually treated as opaque containers for functions. We can run them and collect their activations, but the weights alone rarely provide a readable map from input features to outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show how the weights construct that feature.

A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix factorization for such tables. When a network uses only additions, multiplications, and ratios of polynomials, its weights can be expanded into coefficient tensors whose entries record input interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit is an inspectable model artifact: candidate structure comes from the trained parameters. Softmax, ordinary activations, and square-root normalization obstruct this direct conversion in conventional networks. Bilinear language and vision models suggest that changing the primitives can recover it [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

We study two linked questions:

> *Can the learned weights of a specialist VLA specify an explicit rational tensor program, and can exact weight analysis coexist with strong closed-loop control?*

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Our contributions are:

1. We construct and mechanically certify a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We derive exact weight-based structure through each softmax-free attention layer and an $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
3. We test exact attention subspaces against fixed equal-rank random controls. The weight-derived subspaces win all 36 primary block-head comparisons.
4. We establish capability in a convolutional instantiation of the same rational family, reaching $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds. A three-checkpoint ensemble reaches 96.2% at three times the inference cost.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i \bar{x}_j. \tag{1}$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Updating this pair preserves input-specific rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has $20{,}137{,}352$ learned parameters.

Training uses $66{,}984$ demonstration frames, $40{,}000$ AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a $180°$ camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real $189/200$ checkpoint at runtime and reconstruct representative operations from saved weights.

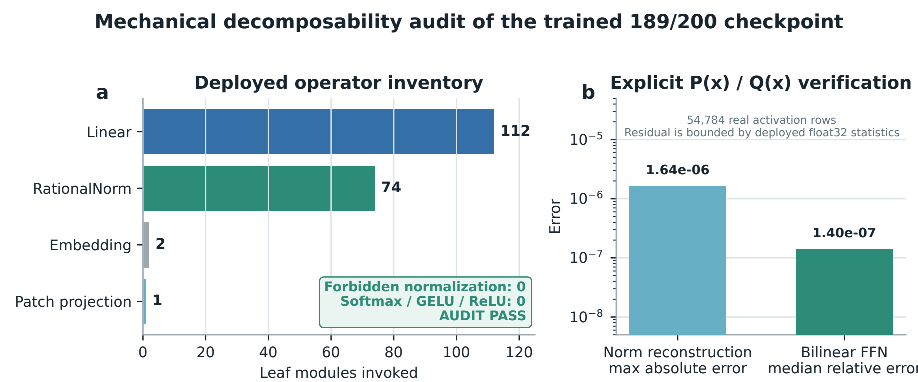

Figure 1: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 The artifact is a capable closed-loop policy

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and $1{,}500$ rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7\% \pm 0.8\%}$
χ-VLA, conv ensemble	$3 \times$ 20.1M	$3 \to 1$	$\mathbf{96.2\%}$
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer's computation.

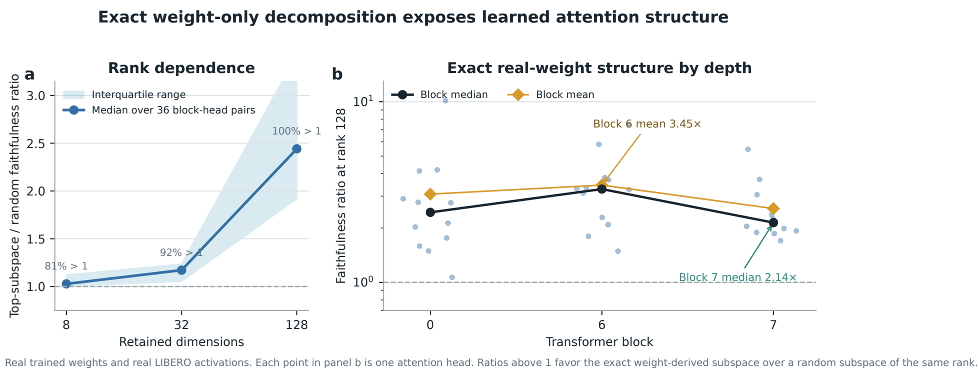

Figure 2: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ-nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The capability result uses convolutional checkpoints, while the exact analysis uses separate ViT checkpoints. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Cache-backed operations join pinned LeRobot task metadata before using official LIBERO task splits. Canonical-state checks fail loudly when requested states are unavailable.

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A Supporting results and decisions

A.1 Per-task ensemble behavior

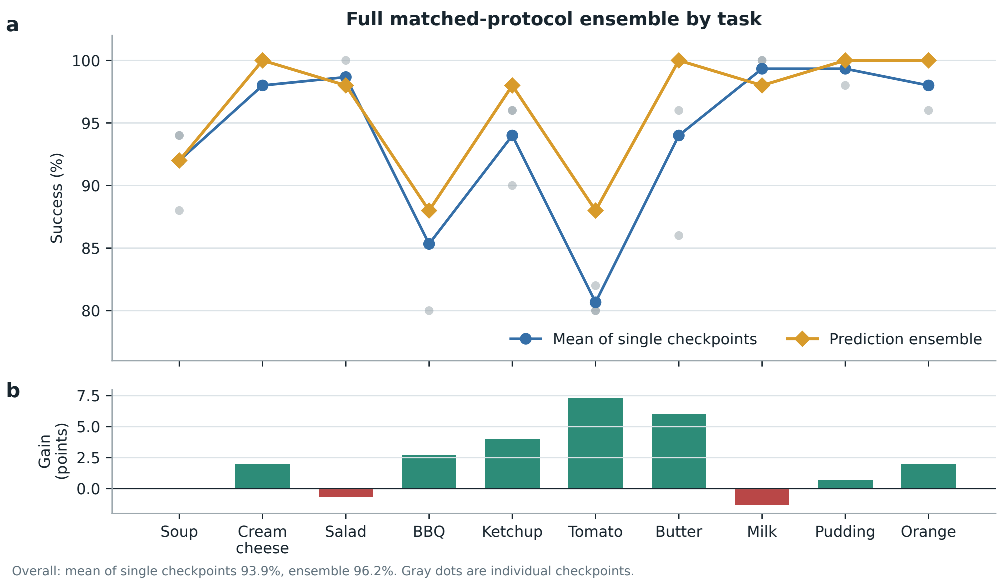

Figure 3: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.2 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.3 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.